



- 2 The Sun may be considered as a uniform sphere with a mass of $1.99 \times 10^{30} \text{ kg}$ and a surface temperature of 5780 K .

A probe with a mass of 2.63 kg moves in a straight line towards the Sun.

When it is at a distance x from the centre of the Sun, the probe measures the gravitational field strength g due to the Sun and the radiant flux intensity F of radiation from the Sun.

- (a) Define gravitational field.

.....
 [1]

- (b) For the position of the probe where $x = 1.47 \times 10^{11} \text{ m}$:

- (i) calculate g

$$g = \dots\dots\dots \text{N kg}^{-1} \quad [2]$$

- (ii) determine the gravitational potential energy E_p of the probe.

$$E_p = \dots\dots\dots \text{J} \quad [2]$$

- (c) (i) Show that, for any particular value of x , the numerical values of g and F are related by

$$g = \frac{4\pi GM}{L} F$$

where M is the mass of the Sun, L is the luminosity of the Sun and G is the gravitational constant.

[3]



(ii) Fig. 2.1 shows the variation of g with F .

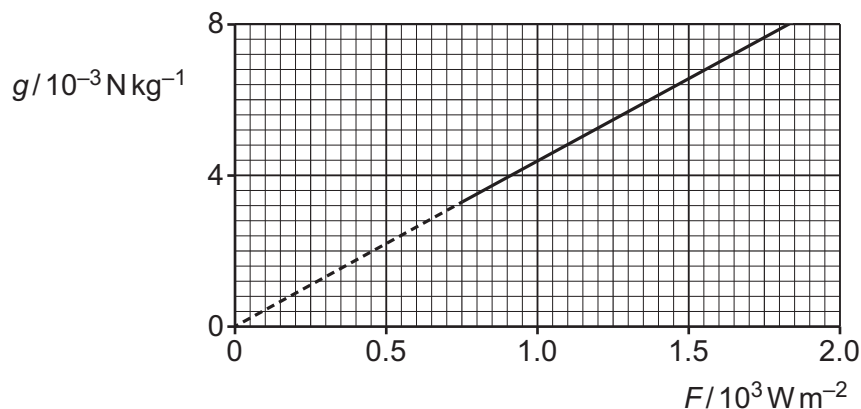


Fig. 2.1

Determine a value for the luminosity L of the Sun. Give a unit with your answer.

$L = \dots\dots\dots$ unit $\dots\dots\dots$ [2]

(iii) Use your answer in (c)(ii) to determine the radius r of the Sun.

$r = \dots\dots\dots$ m [2]

[Total: 12]

